

Meningitis and encephalitis

Meningitis and encephalitis are inflammatory diseases caused mainly by infection. Both can produce symptoms such as sudden fever, a stiff neck, light sensitivity, headaches, drowsiness, vomiting, confusion, and seizures.

Meningitis is an infection of the meninges—the membranes that protect the brain and spinal cord and contain the cerebrospinal fluid that flows throughout the entire nervous system. When infection causes these membranes to swell, the inflammation can ultimately impact every part of the body. Young children whose immune systems are not fully developed are most at risk, although the disease can strike people of any age.

The main cause of meningitis is germs entering the body, whether in the form of bacteria—which can also lead to septicemia, or blood poisoning—viruses, or fungal infections. However, certain drugs,

such as anesthetics, contain substances that can also irritate the meninges, triggering meningitis.

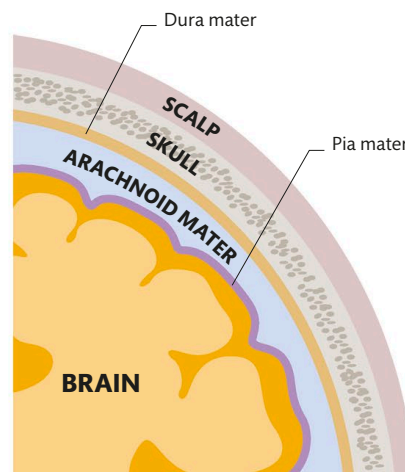
Encephalitis

Encephalitis is an inflammation of the brain itself, due to an infection or to the immune system attacking the brain in error. A person of any age can contract encephalitis, which can cause serious symptoms such as muscle weakness, sudden dementia, loss of consciousness, seizures, and even death.

Sites of infection

The meninges are the outer dura mater, the middle arachnoid, and inner pia mater. In all forms of meningitis, they become inflamed and impair brain function.

1 MILLION
THE NUMBER OF PEOPLE
WORLDWIDE AFFECTED
BY MENINGITIS
EACH YEAR



Brain abscess

Also known as cerebral abscesses, brain abscesses are pus-filled swellings in the brain, which often form after an infection or severe head injury that have allowed bacteria or fungi to enter the brain tissue.

The symptoms of a brain abscess may develop slowly or quickly. They can include symptoms such as a localized headache that cannot be relieved by painkillers, neurological problems such as muscle weakness and slurred speech, changes in mental state, high temperature, seizures, nausea, stiff neck, and changes in vision.

Brain abscesses are usually caused by an infection in another

part of the skull such as an ear infection or sinusitis; an infection in another part of the body—for example, a pneumonia infection spreading via the blood; or trauma, such as a severe head injury that cracks open the skull.

Assessment and diagnoses of brain abscesses are made via blood tests and a CT or MRI scan. Medication and surgery are the most common forms of treatment.

CONGENITAL HEART DEFECT

A brain abscess can also be a rare complication of a group of conditions known as cyanotic heart disease, which are congenital (present at birth). These cause abnormal blood flow through the heart and lungs, allowing poorly oxygenated blood to be pumped around the body. This oxygen-deprived blood gives affected children's skin a blue, or cyanotic, color and severely limits their physical activity.